

Async-RBench

(a) Task setup



Same task
Same starting tasks



Event rules
8 event themes



Revise artifacts



Preserve artifacts

(b) Linear



Subagent A



Subagent B



Subagent C



Wait for all
One batch

Main agent

Combine results

(c) Async



Main agent

Working

Respond

Continue

Replan · Keep



Scenario control

Rule: A first

A

B

Events

Control



Subagent A



Subagent B



Subagent C



Run records

(d) Scoring



Private checks

BTS



Linear

BTS



Async

DRS

Async



Trace + state



½ Process

+

½ Outcome

Equal theme weights